

Demand Forecasting

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Meaning of Forecasting

- In preparing **plans for the future**, the management has to make some predictions about what is likely to happen in the future.
- Forecasting is the process of estimating the relevant events of future, **based on the analysis of their past and present behaviour**.
- Forecasting may be defined as the process of assessing the future normally using calculations and projections that take account of the **past performance, current trends and anticipated changes in the foreseeable period ahead**.

Features of forecasting

- Forecasting relates to **future events**.
- Forecasting is needed for **planning process** because it devises the future course of action.
- It defines the **probability of happening** of future events. Therefore, the happening of future events can be precise only to a certain extent.
- Forecasting is made by **analyzing the past and present factors** which are relevant for the functioning of an organisation.
- The analysis of various factors may require the use of statistical and mathematical tools and techniques.

Role of Forecasting

- Since planning involves the future, no usable plan can be made unless the manager is able to take all possible future events into account.
- Forecasting is a critical element in the planning process.
- In fact, every decision in the organisation is based on some sort of forecasting.

Forecasting

- An estimate of sales for a specified future period
 - under proposed marketing plan ,
 - economic and other forces external to the organization.

?? Before....Investment decision

- What should be the size or amount of capital required?
- How large should be the size of the workforce?
- What should be the size of the order and safety stock?
- What should be the capacity of the plant?

Prediction- vs -Forecasting

- Prediction is an estimate of future event through subjective consideration. It is based on managers skill, experience and judgement.
- Forecasting is objective and an estimate of future event based on the historical data. It requires statistical and management science techniques.

Need for demand forecasting

- Activities depends on future sales
- Future expansion
- Utilisation of plant capacity
- Material planning
- Planning product mix
- Product diversification or modification
- Research and development
- Uncertainty in the market-(techno-economic)

Steps in forecasting process

- Determine the purpose
- Select the time horizon (period) and data source (company records, Journals, govt.publications, newspapers etc)
- Determine the forecasting methods- based on past data,
required accuracy, cost etc
- Make the forecast

Long term forecast-Above 5 years

- Capacity planning
- Investment planning
- Product diversification
- Sales and advertising budgets

Short term forecasting----Less than one year

- Material control
- Loading
- scheduling

Factors affecting Demand Forecast

- Business cycle- seasonal patterns
- Customers plan
- Product life cycle
- Competition and price
- Quality of product
- Credit policy
- Design of goods
- Service reputation
- Advertising and sales effort

Forecasting methods-classification

- Judgemental Techniques
- Time series methods
- Causal methods (econometric forecasting)

Judgemental Techniques-

Relies on the art of human judgement

- Opinion survey-buyers, executives, customers and distributors
- Marketing trials
- Market research
- Delphi Technique

Time series analysis

- **Past data** arranged in a chronological order as a dependent variable and **time** as an independent variable.
- Determines the trend underlying the demand and extrapolate the future trend.
- Does not study the factors that influence the demand.
- All the factors that shape the demand are grouped into one factor time.

Time series analysis-methods

- Least square method of forecasting
- Moving average method
- Weighted moving average method
- Exponential smoothing method

Causal forecasting methods

- Correlation- is determining the degree of closeness of the relationship between two variables (coefficient of correlation – value ranges between +1 and -1) .Positive and negative correlation.
- Regression-Establishes the relationship between variables. The coefficient of regression gives the value by which one variable changes (dependent variable) when there is a change in other variable (independent variable).

Least square method

- Line of best fit
- Sum of the squares of the deviation of the various points from the line of best fit is minimum or least.
- $y=a+bx$; a is the y -intercept; b is the slope of the line.
- $a=\Sigma y/N$; $b=\Sigma xy/\Sigma x^2$

Problem 1

- The sales data for a product is given below. Forecast the demand for the next 2 years using the least square method.

year	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Sales (in lakhs)	30	33	37	39	42	46	48	50	55	58

Solution- problem1

year	Sales(y)	Deviation(x)	x^2	xy
2011	30	-4.5		
2012	33	-3.5		
2013	37	-2.5		
2014	39	-1.5		
2015	42	-0.5		
2016	46	+0.5		
2017	48	1.5		
2018	50	2.5		
2019	55	3.5		
2020	58	4.5		
N=10	$\Sigma y =$	$\Sigma x = 0$	$\Sigma x^2 = 82.5$	$\Sigma xy = 251$

Solution- problem2

year	Sales(y)	Deviation(x)	x^2	xy
2012	33	-4	16	
2013	37	-3	9	
2014	39	-2	4	
2015	42	-1	1	
2016	46	0	0	
2017	48	1	1	
2018	50	2	4	
2019	55	3	9	
2020	58	4	16	
N=9	$\Sigma y=408$	$\Sigma x=0$	$\Sigma x^2=60$	$\Sigma xy=182$

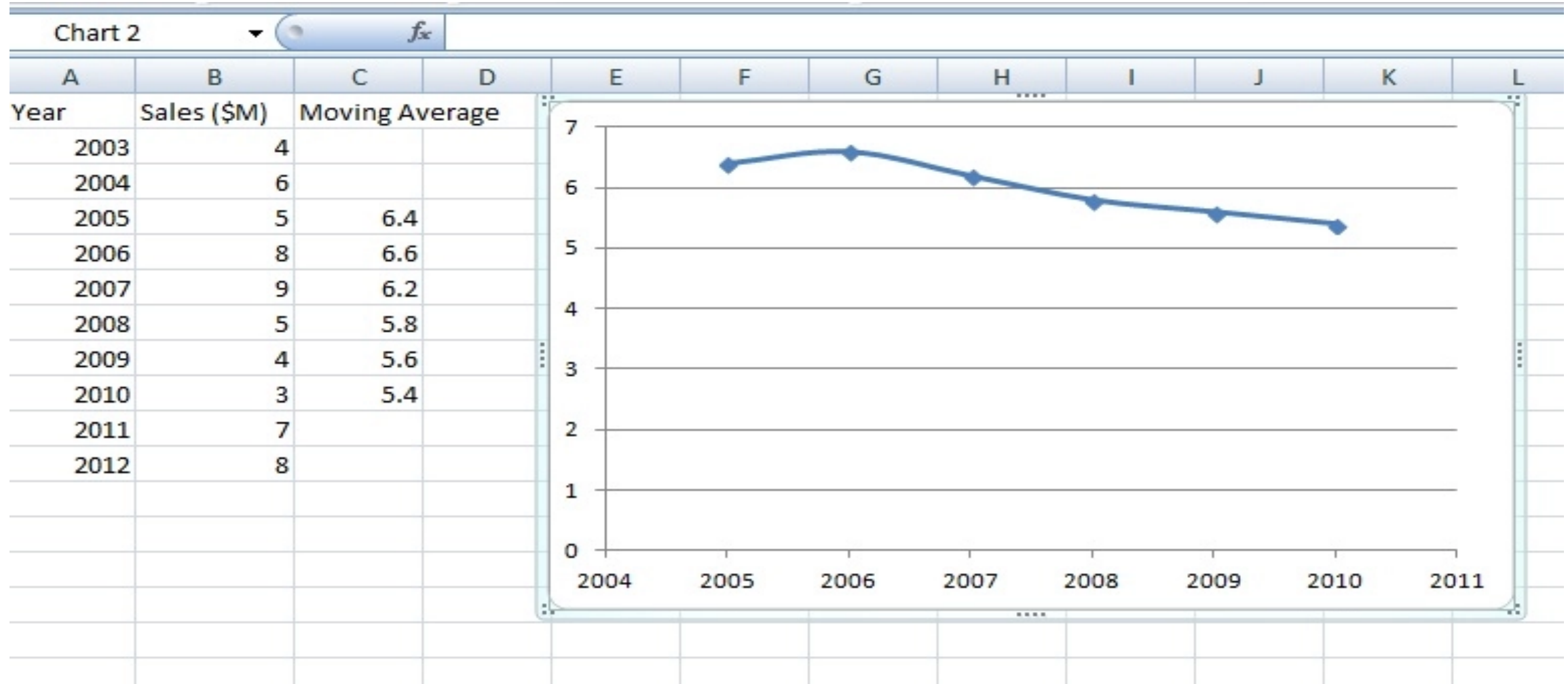
problem1

- $a = 438/10 = 43.8$
- $b = 3.04$
- $Y = a + bx$
- For 2021, $x = 5.5$; $y = 43.8 + 3.04 * 5.5 = 60.52$
- For 2022, $x = 6.5$; $y =$

Moving average method

- It's a **technique** to get an overall idea of the trends in a data set
- It is an **average** of any subset of numbers.
- The **moving average** is extremely useful for **forecasting** long-term trends.
- We can calculate it for any period of time.

Moving average- (5 year)



Simple 3 months moving average-Problem 1- to find the demand forecast for the month of Oct. (Ans.86)

Time	Demand (D)	Moving average	Forecast (F)	Error (D-F)	Absolute error= (D-F)/D
Jan	95				
Feb	100	94			
Mar	87	103			
Apr	123	100	94		
May	90	103	103		
June	96	87	100		
July	75	83	103		
Aug	78	86	87		
Sep	106		83		

Weighted Moving Average

Forecasting Methods

- Weighted Moving Average Example
- Assume $n = 2$, $W_1 = 0.7$, $W_2 = 0.3$

Week	Demand	
1	125	
2	175	
3	150	$(0.7)(175) + (0.3)(125) = 160$
4	150	$(0.7)(150) + (0.3)(175) = 157.5$
5	160	$(0.7)(150) + (0.3)(150) = 150$
		$(0.7)(160) + (0.3)(150) = 157$

Problem-2

- The past 6 months data on the sales of a company is given below, compute 3 months moving average forecast for the month of December 2020. Also compute 3 months weighted moving average for the month of December 2020, where the weights are 0.5 for the latest month, 0.3 and 0.2 for other months respectively.

Month	Sales (in Lakhs Rs)
June 2020	585
July	610
August	675
September	750
October	860
November	970

Exponential smoothing method

Demand for the most recent data is given more weightage and the weights assigned to older periods decreases exponentially

This method requires only the current demand and the forecasted demand for the current month.

$$F_t = F_{t-1} + \alpha [D_{t-1} - F_{t-1}]$$

Where α is the smoothing constant.

value for α is small if the demand pattern is smooth (0.05- 0.1) and is large up to 0.7 if the demand pattern is fluctuating

Problem 3

- The demand for a product are 300 and 350 units for the months of September and October respectively. Using 200 units as the forecast for September, compute the forecast for the month of November? Assume the smoothing constant $\alpha = 0.7$

$$D_{t-1} = 300 \dots (\text{September})$$

$$F_{t-1} = 200 \dots (\text{September})$$